

Regular Expressions and Languages:-

Regular Languages:-

Regular Language is a Set accepted by finite automaton.

Regular Expressions:-

Regular Expressions are the type of Languages defining notation. The regular Expressions are used to describe the regular Language. The regular Expression operations are union, Concatenation and Closure. Regular Expressions are the Languages accepted by finite automata. Each regular expressions 'r' denotes a Language " $L(r)$ ".

1. ϕ is a regular expression that denotes the empty set $\{\}$.
2. ϵ is a regular Expression that denotes $\{\epsilon\}$ that is the Set containing the empty string.
3. If 'a' is a Symbol Present in Σ then 'a' is a regular expression that denotes $\{a\}$ that is the Set containing the string 'a'.

4. If 'r' and 's' are the regular expressions denoting the regular languages $L(r)$ and $L(s)$ then,

* $(r) + (s)$ is the regular expression denoting $L(r) \cup L(s)$.

* $(r) \cdot (s)$ is the regular expression denoting $L(r) \cdot L(s)$.

* $(r)^*$ is the regular expression denoting $(L(r))^*$.

Operators of Regular Expression:-

* There are 3 operations on languages.

1. Union
2. Concatenation.
3. Closure.

1. Union:-

Union of 2 languages L_1 and L_2 ($L_1 \cup L_2$) is the set of strings that either in L_1 or L_2 or both $L_1 \cup L_2$.

$$L_1 \cup L_2 = \{x \text{ or } y; x \text{ is in } L_1, \text{ or } y \text{ is in } L_2\}$$

$$L_1 = \{0, 11, 01\} \quad L_2 = \{\epsilon, 00, 01\}$$

$$L_1 \cup L_2 = \{0, 11, 01, \epsilon, 00, 01\}$$

$$L_1 \cup L_2 = \{0, 11, 01, \epsilon, 00\}$$

2. Concatenation:-

The Concatenation of 2 Languages L_1 and L_2 is represented as $(L_1 \cdot L_2)$ which is formed by choosing a string in L_1 and followed by a string in L_2 in all possible combination.

$$L_1 \cdot L_2 = \{xy; x \text{ is in } L_1, y \text{ is in } L_2\}$$

$$L_1 = \{10, 1\} \quad L_2 = \{011, 11\}$$

$$L_1 \cdot L_2 = \{10011, 1011, 1011, 1111\}$$

$$L_1 \cdot L_2 = \{10011, 1011, 1111\}$$

3. Closure:-

The Kleene Closure of a language L denoted as L^* is defined as a set of strings that can be formed by taking any number of strings from L .

$$L^* = \bigcup_{i=0}^{\infty} L^i$$

$$L^0 = \{\epsilon\} \quad L^i = \{L \cdot L^{i-1}\} \quad i \geq 1$$

Positive closure:-

$$L^+ = \bigcup_{i=1}^{\infty} L^i \quad \therefore \text{Here no } \epsilon$$

$$L = \{10, 1\}$$

$$L^+ = \{\epsilon, 10, 1, 1010, 11\}$$

$$L^+ = \{10, 1, 1010, 11\}$$

Problems on Regular Expressions:-

1. Set of all strings of zero or more a's & b

$$R.E :- a^* b^*$$

$$R.L = \{\epsilon, a, aa, aaa, \dots\}$$

2. Set of all strings with zero or more instances of a or b & ,

$$R.E :- (a+b)^*$$

$$R.L = \{\epsilon, a, aa, b, bab, abb, \dots\}$$

3. Set of all strings over $\{a, b, c\}$ with any number of a's followed by any number of b's followed by any number of c's.

$$R.E :- a^* b^* c^*$$

$$R.L = \{\epsilon, a, b, c, abc, aabbcc, \dots\}$$

4. Set of all strings over $\{a, b, c\}$ with at least one 'a' followed by at least one 'b' followed by at least one 'c'

$$R.E :- a^+ b^+ c^+$$

$$R.L :- \{abc, aabc, aabbc, \dots\}$$

5. Set of strings of a's and b's of length 2 or more

$$R.E :- (a+b)^2$$

$$R.L :- \{aa, ab, ba, bb\}$$

6. Set of all strings over $\{0, 1\}$ starting and ending with 0

$$R.E :- 0(0+1)^* 0$$

$$R.L :- \{00, 000, 010, 0010, \dots\}$$

7. Set of all strings over $\{a, b\}$ ending with aba

aba is

$$R.E :- (a+b)^* aba$$

$$R.L :- \{aba, aaba, baba, ababa, \dots\}$$

8. Set of all strings over $\{a, b, c\}$ containing at least one 'a' and at least one 'b'

$$R.E :- (a+b+c)^* a (a+b+c)^* b (a+b+c)^*$$

$$R.L :- \{ab, aab, abb, bab, \dots\}$$

9. Set of strings over $\{1\}$ having odd length of strings is

$$R.E :- 1(11)^*$$

$$R.L :- \{1, 111, 11111, \dots\}$$

10. Set of all strings over $\{0,1\}$ ending with 00 is

$$R.E :- (0+1)^* 00$$

$$R.L :- \{00, 000, 100, 0100, \dots\}$$

11. Set of all strings over $\{0,1\}$ beginning with 00 and ending with 1 is

$$R.E :- 00(0+1)^* 1$$

$$R.L :- \{001, 0001, 0011, 00001, \dots\}$$

12. Set of all strings over $\{0,1\}$ with at most one pair of 3 consecutive 1's is

$$R.E :- (0+1)^* 111 (0+1)^*$$

$$R.L :- \{111, 0111, 11110, \dots\}$$

13. Set of all strings with even number of 1's over $\{0,1\}$ is

$$R.E :- (0^* 1 0^* 1 0^*)^*$$

$$R.L :- \{11, 0111, 0110, \dots\}$$

14. Set of all strings over $\{0,1\}$ where no. of 0's is divisible by 5 is,

$$R.E :- 1^*(00000)^*1^*$$

$$R.L :- \{ \epsilon, 100000, 000001, 11000001, \dots \}$$

15. Set of all strings over $\{a,b,c\}$ which contains at least one 'a' is

$$R.E :- (b+c)^* a (b+c)^*$$

$$R.L :- \{ a, ba, ca, bac, bbac, \dots \}$$

16. Give the regular expression for the language $\{b^2, b^5, b^8, \dots\}$

$$R.E :- bb(bbb)^*$$

17. Give the regular expression for the language $\{a, bb, aa, abb, ba, bbb, \dots\}$

$$R.E :- (a+b)^*(a+bb)$$

18. Set of all strings over $\{a,b,c\}$ in which all runs of a's have length that are multiples of 3 is

$$R.E :- (aaa+b+c)^*$$

$$R.L :- \{ \epsilon, aaa, aaabac, aaabaaac, \dots \}$$

19. Set of all strings $\{0, 1\}$ that have even length is

$$R.E: (00+01+10+11)^*$$

$$R.L: \{\epsilon, 0001, 1011, 0011, 000110, \dots\}$$

20. Set of all strings over $\{0, 1\}$ that have odd length is

$$R.E: (0+1)(00+10+01+11)^*$$

$$R.L: \{0, 1, 000, 11001, 01101, \dots\}$$

21. Set of all strings over $\{0, 1\}$ in which 3rd symbol from right of string is 1,

$$R.E: (0+1)^*1(0+1)^*$$

$$R.L: \{101, 100, 110, 111, 1101, 00100, \dots\}$$

22. Set of all strings over $\{0, 1\}$ starting

with 1 and not having 2 consecutive 0's is

$$R.E: (1+10)^+$$

$$R.L: \{110, 101, 11010, \dots\}$$

Start with zero, no. of consecutive 1's

$$R.E: (0+01)^+$$

Q3. Give the regular expression for the language
 $\{a, c, ab, cb, abb, cbb, abbb, \dots\}$

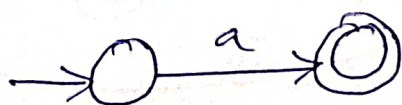
R.E:- $(a+c)b^*$.

Converting Regular Expressions to Automata-

The regular expressions can be converted to NFA with ϵ -Transition.

a. First automata is constructed for the basic expressions: single symbols, ϵ & ϕ

* Single symbol transition.



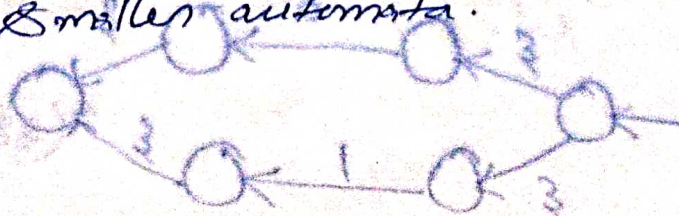
* ϵ -Transition.



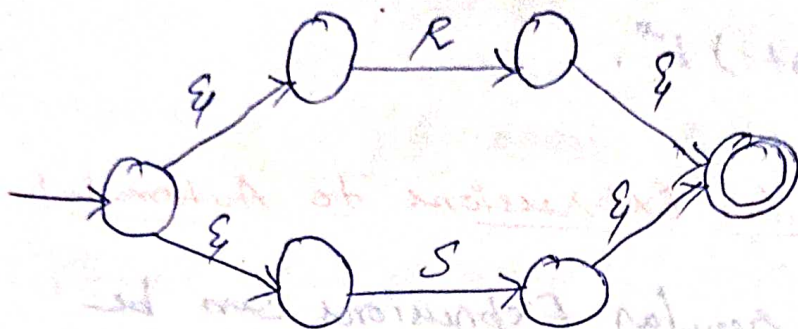
* ϕ -Transition



b. Then these automata are combined into larger automata that accept the union, concatenation or closure of the language accepted by smaller automata.



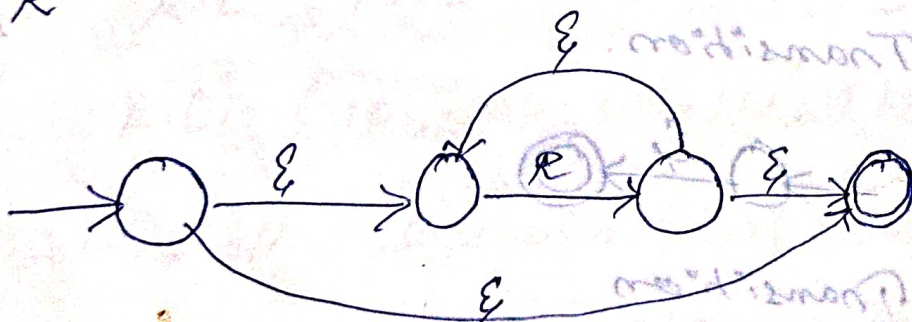
1. $R+S$



2. $R \cdot S$



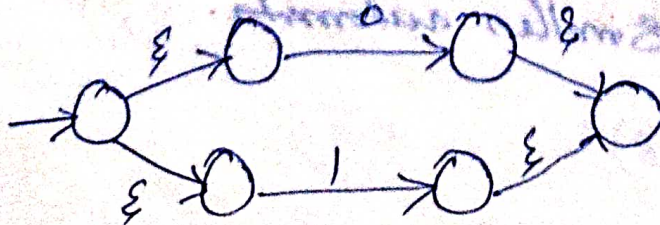
3. R^*



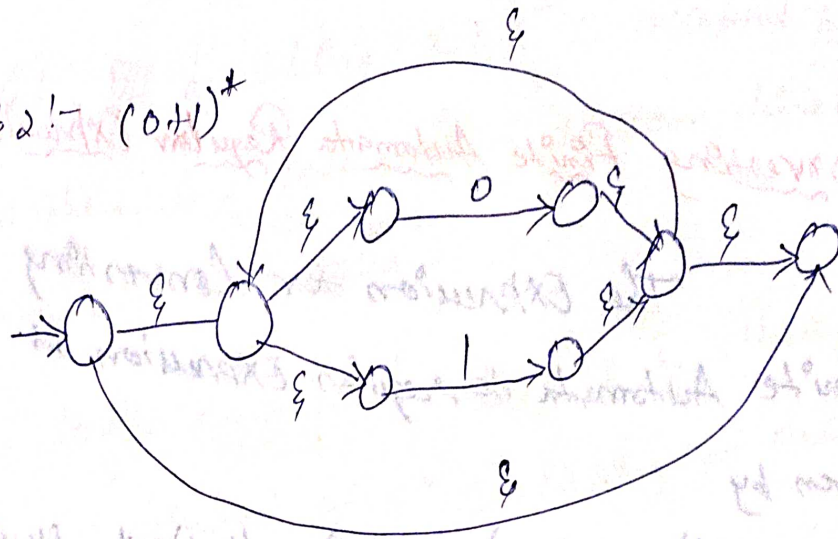
Ex-1

1. Convert the regular expression $(1+0)^*$ to NFA with ϵ -Transition.

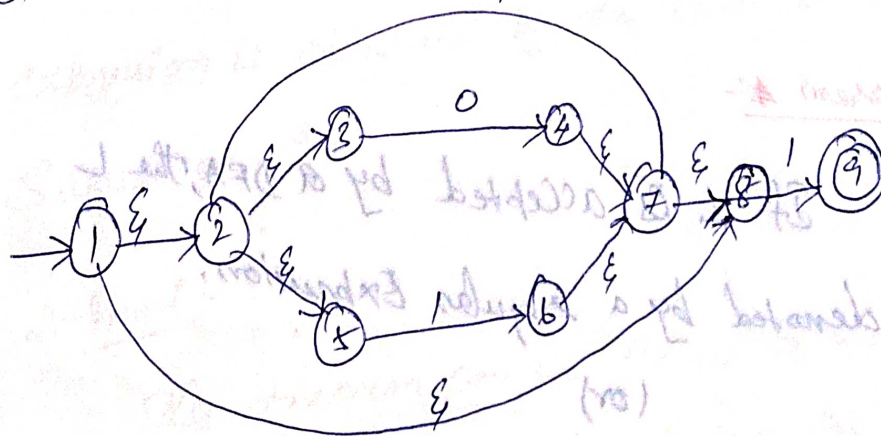
Step 1:- $(1+0)$



Step 2:- $(0+1)^*$



Step 3:- $(0+1)^*1$



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2. Convert the regular Expression $(01+101)$ to NFA with ϵ -transition.

3. Convert the regular Expression $(01+1)^*$ to NFA with ϵ -transition.

4. Convert the regular Expression $011(0+1)^*$ to NFA with ϵ -transition.

5. $(ab+ca)^*b$

6. $(a+ab)^*(a+bb)^*$

Converting Finite Automata Regular Expressions

The Expression for converting Finite Automata to regular Expression is given by

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} \cdot (R_{kk}^{(k-1)})^* \cdot R_{kj}^{(k-1)}$$

Theorem 4:-

If L is accepted by a DFA, the L is denoted by a regular Expression.

(or)

If $L = L(A)$ for some DFA A , then there is a regular expression R such that

$$L = L(R)$$

or

Any set L accepted by a finite automata M is represented by a regular Expression.

1. Given :-

Let the DFA

$$M = (\{q_1, q_2, \dots, q_n\}, \Sigma, \delta, q_1, F)$$

Let. If a string $x \in \Sigma^*$ is accepted by M , then there is a path from q_1 to some final state with path value x .

So to each final state, say q_j there exists a subset of Σ^* consisting of paths from q_1 to q_j . Hence, the required regular expression is obtained by the union of such subsets of Σ^* .

2. Proof:-

This theorem can be proved by the Principle of Mathematical Induction for $R_{ij}^{(k)}$.

3. To Prove:-

The main part of the proof lies in the construction of subsets of paths from the state q_i to the state q_j .

Let $R_{ij}^{(k)}$ denote the set of path values of paths from q_i to q_j whose intermediate vertices lie in $\{q_1, \dots, q_k\}$.

We construct $R_{ij}^{(k)}$ for $k = 0, \dots, n$

$L(A) \cup R_{ij}^{(n)}$ recursively as follows.

4. Basic step: ($k=0$)

If $k=0$, there will be no intermediate values between i and j .

$\therefore R_{ij}^{(0)}$ can be described as follows.

Case 1:- ($i=j$)



$$R_{ij}^{(0)} = R_{ii}^{(0)} = \{a / \delta(r_i, a) = r_i \cup \{\epsilon\}\}.$$

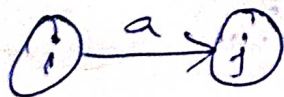
denotes the set of all symbols a such that a or ϵ .

$$R_{ij}^{(0)} = R_{ii}^{(0)} = a + \epsilon$$

where a is the set of all

Symbol from Σ i.e. $a_1, \dots, a_k$.

Case 2:- ($i \neq j$)



R_{ij} denotes the set of all symbols from i to j

$$R_{ij}^{(0)} = \{a / \delta(r_i, a) = r_j\}$$

$$(i.e.) R_{ij}^{(0)} = a_1 + a_2 + a_3 + \dots + a_k$$

Case 2:-
If there is no any direct arc from i to j then $R_{ij}^{(0)} = \emptyset$.

5. Induction Hypothesis:- ($k \geq 1$)

Assume that the theorem is true for $k=1$ then we must prove that is also true for $k=n$.

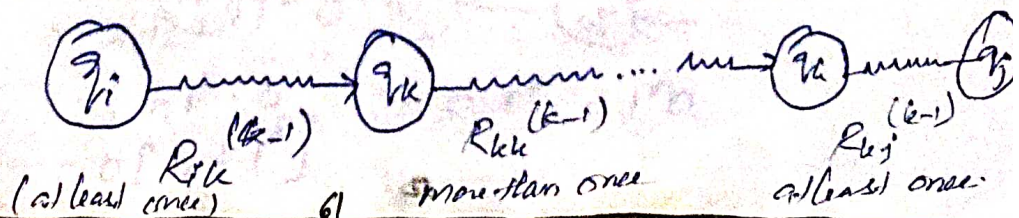
6. Proof at Induction Step:-

Consider a path from q_i to q_j where intermediate vertices lie in $\{q_1, \dots, q_n\}$, then the R.E can be constructed as follows.

(i) Suppose if the path does not pass through q_k , then its path should be

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} \quad \text{--- (1)}$$

(ii) Otherwise, if the path goes through q_k at least once or more than once then the path can be split into several paths as follows.



The Regular Expression for the above Set of all paths from i to j is:

$$R_{ik}^{(k-1)} \cdot (R_{kk}^{(k-1)})^* \cdot R_{kj}^{(k-1)} \quad \text{--- (2)}$$

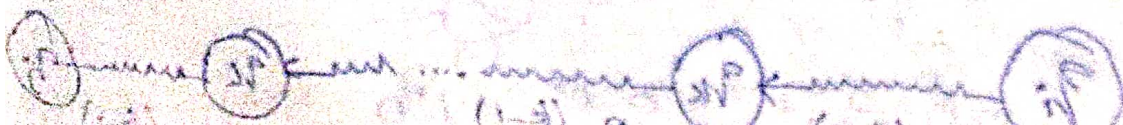
By combining equation (1) & (2) we get the R.E for $R_{ij}^{(k)}$

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} + R_{ik}^{(k-1)} \cdot (R_{kk}^{(k-1)})^* \cdot R_{kj}^{(k-1)}$$

6. Conclusion:-

Therefore, we conclude that the R.E $R_{ij}^{(n)}$ can be constructed for all values of i, j and n .

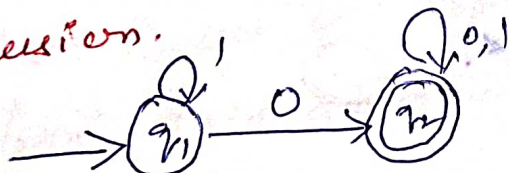
As $Q = \{q_1, \dots, q_n\}$, the R.E $R_{ij}^{(n)}$ denotes the sum of all the paths from q_1 to q_j such that q_1 is initial state & q_j is some final state.



Formulas:-

1. $(\epsilon + 1)^* = 1^*$
2. $1^* (\epsilon + 1)^* = 1^*$
3. $(\epsilon + 1) + 1^* = 1^*$
4. $0 + 1^* 0 = 1^* 0$
5. $\epsilon + 00^* = 0^*$
6. $(1 + 0)^* = (1^* 0^*)^*$
7. $\phi \cdot 0 = \phi$
8. $\phi + 0 = 0, \phi^* = \epsilon$
9. $\phi^* = \epsilon$
10. $\epsilon \cdot 0 = 0$
11. $\epsilon + 0 = 0$ or $\epsilon + 1 = 1$
12. $1 1^* = 1^* 1 = 1^*$
13. $0 0^* = 0^*$
14. $R R^* = R^+ = R^*$
15. $\epsilon + R^+ = R^*$
16. $R + R^* = R^*$

1. Convert the following DFA to regular Expression.



$$R_{11}^{(1)} = 1 + \epsilon$$

$$R_{12}^{(1)} = 0$$

$$R_{21}^{(1)} = \phi$$

$$R_{22}^{(1)} = 1 + 0 + \epsilon$$

$$R_{12}^{(2)} = R_{12}^{(1)} + R_{11}^{(1)} \cdot R_{22}^{(1)} \cdot R_{12}^{(1)}$$

$$R_{12}^{(1)} = R_{12}^{(1)} + R_{11}^{(1)} \cdot R_{11}^{(1)*} \cdot R_{12}^{(1)}$$

$$= 0 + (1 + \epsilon) \cdot (1 + \epsilon)^* \cdot 0$$

$$= 0 + 1 \cdot 1^* \cdot 0$$

$$= 0 + 1^* \cdot 0$$

$$= 0[\epsilon + 1^*]$$

$$= 0 1^* = 1^* 0$$

$$R_{22}^{(1)} = R_{22}^{(0)} + R_{21}^{(0)} \cdot R_{11}^{(b)*} \cdot R_{12}^{(0)}$$

$$= (1+0+\epsilon) + \phi \cdot (1+\epsilon)^* \cdot 0$$

$$= (1+0+\epsilon) + \phi$$

$$= 1+0$$

$$R_{12}^{(2)} = R_{12}^{(1)} + R_{12}^{(1)} \cdot R_{22}^{(1)*} \cdot R_{22}^{(1)}$$

$$= 1^*0 + 1^*0 \cdot (1+0)^* (1+0)$$

$$= 1^*0 + 1^*0 \cdot (1+0)^*$$

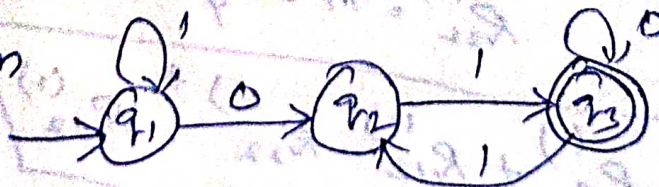
$$= 1^*0 [\epsilon + (1+0)^*]$$

$$= 1^*0 (1+0)^*$$

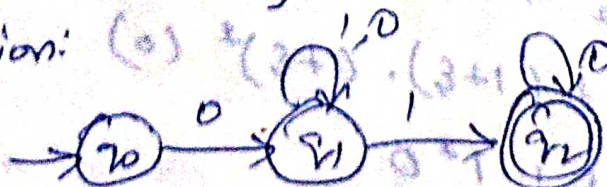
The regular expression for the above DFA is $1^*0(1+0)^*$.

H/w:-

1. Convert the following DFA to regular Expression

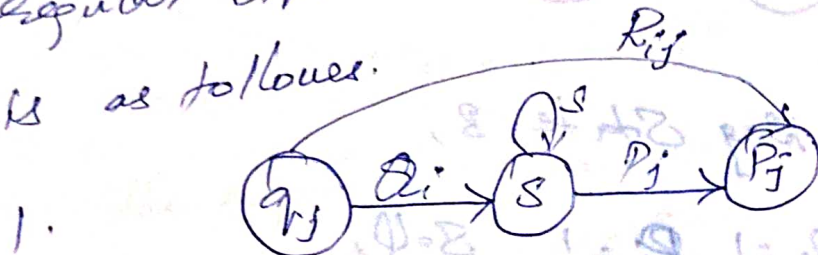


2. Convert the following DFA to regular Expression:

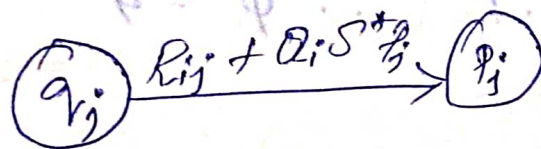


Converting DFA's to Regular Expression by eliminating states

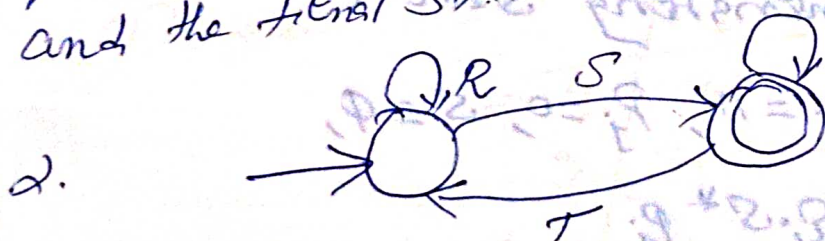
The Strategy for Constructing a regular Expression from a finite automaton is as follows.



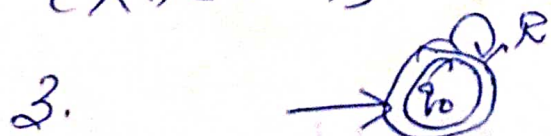
After eliminating state S, the regular expression is $R_{ij} + Q_i S^* P_j$.



For each state q, the above reduction process is applied to get an equivalent automaton with the start state and the final state.

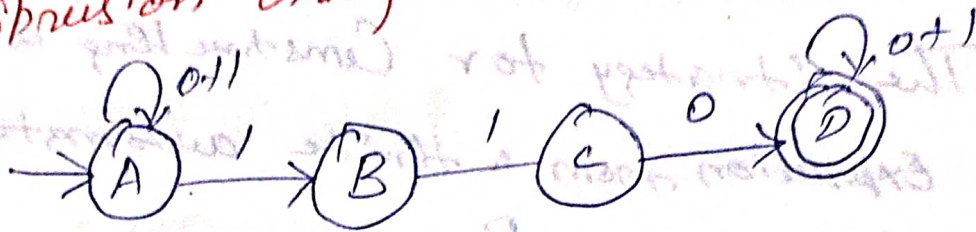


The regular expression is $(R + S U^* T)^* S U^*$.



At the start state and the final state are same means, the regular expression is R^* .

1. Convert the following NFA to regular Expression using state elimination.



Step 1:- Eliminating State B,

$$R_{ij} = \emptyset, Q_i = 1, P_j = 1, S = \emptyset,$$

$$= R_{ij} + Q_i \cdot S^* \cdot P_j$$

$$= \emptyset + 1 \cdot \emptyset^* \cdot 1 \quad \therefore \emptyset^* = \epsilon$$

$$= \emptyset + 1 \cdot 1$$

$$= 11$$



Step 2:- Eliminating State C,

$$R_{ij} = \emptyset, Q_i = 11, P_j = 0, S = \emptyset,$$

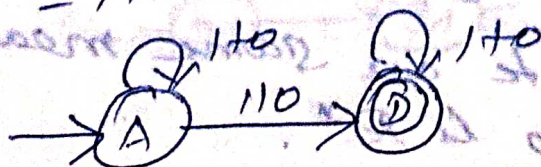
$$= R_{ij} + Q_i \cdot S^* \cdot P_j$$

$$= \emptyset + 11 \cdot \emptyset^* \cdot 0$$

$$= \emptyset + 11 \cdot \epsilon \cdot 0$$

$$= \emptyset + 110$$

$$= 110$$



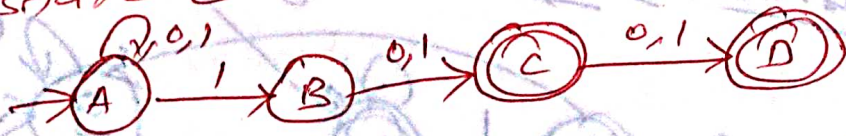
Step 3:- $R = 1 + 0$, $S = 110$, $T = \phi$, $U = 1 + 0$

$$\begin{aligned}
 &= (R + S U^* +)^* S U^* \\
 &= ((1+0) + 110 (1+0)^* \cdot \emptyset)^* \cdot 110 (1+0)^* \\
 &= ((1+0) + \emptyset)^* \cdot 110 (1+0)^* \\
 &= (1+0)^* \cdot 110 (1+0)^*
 \end{aligned}$$

The regular expression for the above NFA is $(1+0)^* 110 (1+0)^*$

H/w:-

H/w:-
1. Convert the following NFA to regular expression
Using state elimination.



2. Convert the following DFA to regular Expression using state elimination.

Transition Table

$$\begin{array}{c|cc}
 & 0 & 1 \\
 \hline
 \rightarrow q_1 & q_2 & q_1 \\
 q_2 & q_3 & q_1 \\
 * q_3 & q_3 & q_2
 \end{array}$$

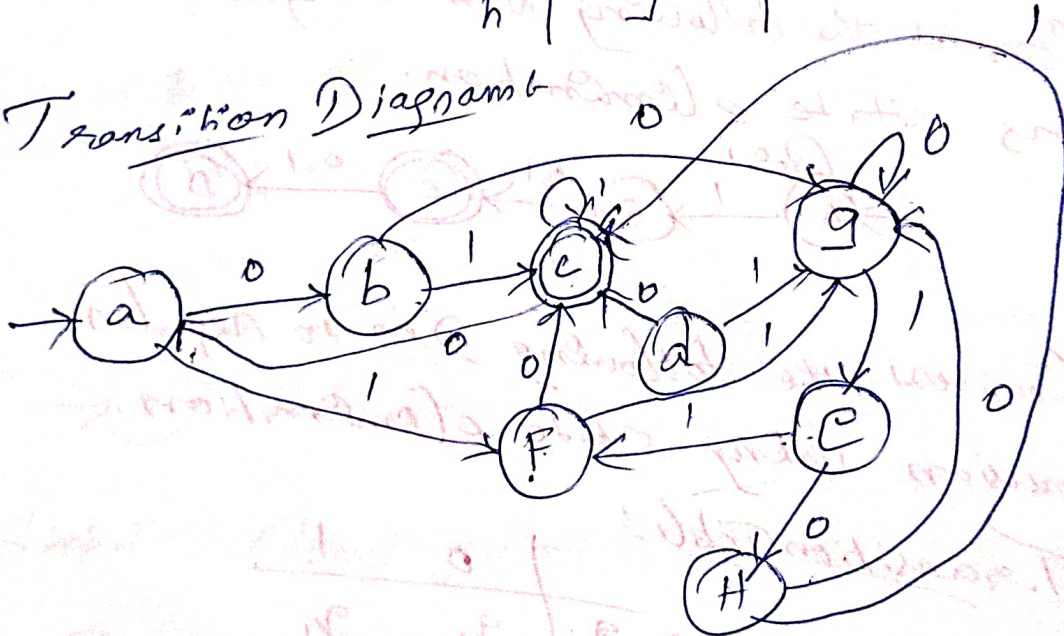
11a to 4.2
[4.2]

Equivalence and Minimization of Automata

Transition Table:-

	0	1
→ a	b	F
b	g	e
* c	a	c
d	e	g
e	h	f
f	e	g
g	g	e
h	g	c

Transition Diagram

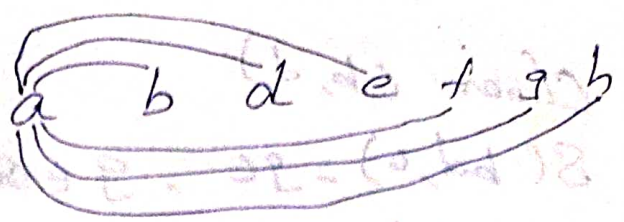


$$Q_0 = \{a, b, c, d, e, f, g, h\}.$$

[Set of all states]

$$Q_1 = \{c\} \rightarrow \text{final state}.$$

$$Q_2 = Q_0 - Q_1 = \{a, b, d, e, f, g, h\}.$$



Steps :-

Choose (a, b)

$$S(ab, 0) = S(a, 0) \cup S(b, 0) = \{b, g\}, \quad b \in Q_2, \quad g \in Q_2$$

$$S(ab, 1) = S(a, 1) \cup S(b, 1) = \{f, c\}, \quad f \in Q_1, \quad c \in Q_1$$

(a, b) is not equivalent.

Choose (a, d)

$$S(ad, 0) = \{b, c\}, \quad b \in Q_2, \quad c \in Q_1, \quad (a, d) \text{ is not equivalent.}$$

Choose (a, e)

$$S(ae, 0) = \{b, h\}, \quad b \in Q_2, \quad h \in Q_2$$

$$S(ae, 1) = \{f, f\}, \quad f \in Q_2$$

(a, e) is equivalent.

Choose (a, f)

$$S(af, 0) = \{b, c\}, \quad b \in Q_2, \quad c \in Q_1, \quad (a, f) \text{ is not equivalent.}$$

Choose (a, g)

$$S(ag, 0) = \{b, g\}, \quad b \in Q_2, \quad g \in Q_2$$

$$S(ag, 1) = \{f, e\}, \quad f \in Q_2, \quad e \in Q_2$$

(a, g) is equivalent.

Choose (a, h)

$$S(ah, 0) = \{b, g\}, \quad b \in Q_2, \quad g \in Q_2$$

$$S(ah, 1) = \{f, c\}, \quad f \in Q_2, \quad c \in Q_1$$

ah is not equivalent.

Choose (b, d)

a b d e f g h

$$S(bd, 0) = gc, g \in Q_2, c \in Q_1$$

(b, d) is not equivalent.

Choose (b, e)

$$S(be, 0) = gh, g \in Q_2, h \in Q_2$$

$$S(be, 1) = cf, c \in Q_1, f \in Q_2$$

(b, e) is not equivalent.

Choose (b, f)

$$S(bf, 0) = gc, g \in Q_2, c \in Q_1$$

(b, f) is not equivalent.

Choose (b, g)

$$S(bg, 0) = gg, g \in Q_2$$

$$S(bg, 1) = ce, c \in Q_1, e \in Q_2$$

(b, g) is not equivalent.

Choose (b, h)

$$S(bh, 0) = gg, g \in Q_2$$

$$S(bh, 1) = cc, c \in Q_1$$

(b, h) is equivalent.

Choose (d, e)

d e f g h

$$S(de, 0) = ch, c \in Q_1, h \in Q_2$$

(d, e) is not equivalent.

Choose (d, f)

$$g(df, 0) = cc, \quad c \in \mathbb{Q}_1$$

$$g(df, 1) = gg, \quad g \in \mathbb{Q}_2$$

(d, f) is equivalent.

Choose (d, g)

$$g(dg, 0) = cg, \quad c \in \mathbb{Q}_1, \quad g \in \mathbb{Q}_2$$

(d, g) is not equivalent.

Choose (d, h)

$$g(dh, 0) = cg, \quad c \in \mathbb{Q}_1, \quad g \in \mathbb{Q}_2$$

(d, h) is not equivalent.

Choose (e, f)

$$g(ef, 0) = hc, \quad h \in \mathbb{Q}_2, \quad c \in \mathbb{Q}_1$$

(e, f) is not equivalent.

Choose (e, g)

$$g(eg, 0) = hg, \quad h \in \mathbb{Q}_2, \quad g \in \mathbb{Q}_2$$

$$g(eg, 1) = fe, \quad f \in \mathbb{Q}_2, \quad e \in \mathbb{Q}_2$$

(e, g) is equivalent.

Choose (e, h)

$$g(eh, 0) = hg, \quad h \in \mathbb{Q}_2, \quad g \in \mathbb{Q}_2$$

$$g(eh, 1) = fc, \quad f \in \mathbb{Q}_2, \quad c \in \mathbb{Q}_1$$

(e, h) is not equivalent.

Choose (f, g)

$\overline{f g h}$

$$S(fg, 0) = cg, \quad c \in \mathcal{Q}_1, \quad g \in \mathcal{Q}_2$$

(f, g) is not equivalent.

Choose (f, h)

$$S(fh, 0) = cg, \quad c \in \mathcal{Q}_1, \quad g \in \mathcal{Q}_2$$

(f, h) is not equivalent.

$\overline{g h}$

Choose (g, h)

$$S(gh, 0) = gg, \quad g \in \mathcal{Q}_2$$

$$S(gh, 1) = ee, \quad e \in \mathcal{Q}_1, \quad c \in \mathcal{Q}_1$$

(g, h) is not equivalent.

Equivalent Pairs

$\{c\}, \{a, e\}, \{g, g\}, \{b, h\}, \{d, f\}$
 $\{e, g\}$

$$S_1 = \{c\}, \{a, e, g\}, \{b, h\}, \{d, f\}$$

Choose (a, e) as pair.

$$S(ae, 0) = bh, \quad bh \in S_1$$

$$S(ae, 1) = ff, \quad f \in \text{same set}$$

(a, e) is equivalent

Choose (q, g) pair:

$\delta(qg, 0) = bg \in \text{diff pair}$

(q, g) is not equivalent

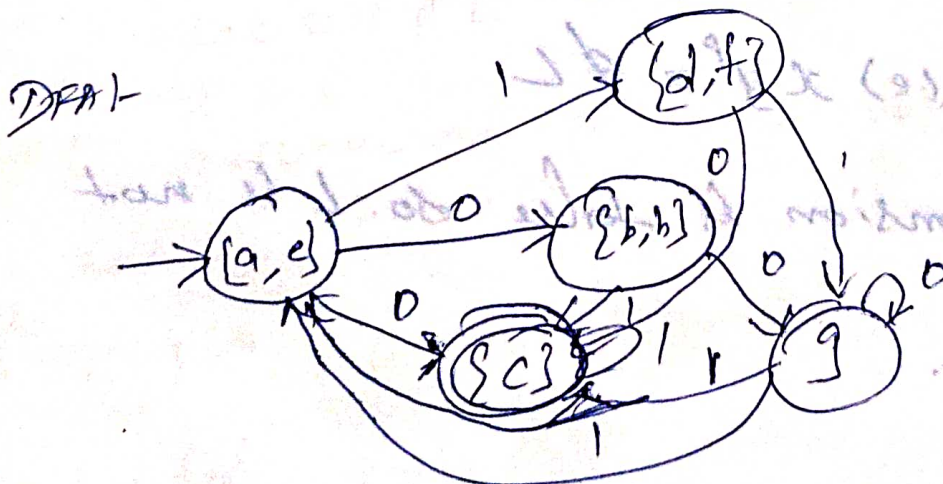
Choose (e, g) pair:

$\delta(eg, 0) = hg \in \text{diff pair}$

(e, g) is not equivalent

$$S_2 = \{c\} \{a, e\} \{g\} \{b, h\} \{d, f\}$$

	0	1		0	1
* $\{c\}$	$\{a, e\}$	$\{c\}$	$\rightarrow a$	b	d
$\rightarrow \{a, c\}$	$\{b, h\}$	$\{d, f\}$	b	g	c
$\{g\}$	$\{g\}$	$\{a, e\}$	* c	a	c
$\{b, h\}$	$\{g\}$	$\{c\}$	d	c	g
$\{d, f\}$	$\{c\}$	$\{g\}$	g	g	a



Pumping Lemma

Proof by Contradiction is used.

Steps:

1. Assume L is regular and also assume n .

2. Choose the string ' w ' (word in L)

Such that $|w| \geq n$, and then decompose
 $\hookrightarrow$ Length of String.

the string ' w ' to xyz . Such that

$|xy| \leq n$ & $|y| > 0$ or $y \neq \epsilon$.

3. Choose a suitable integer i

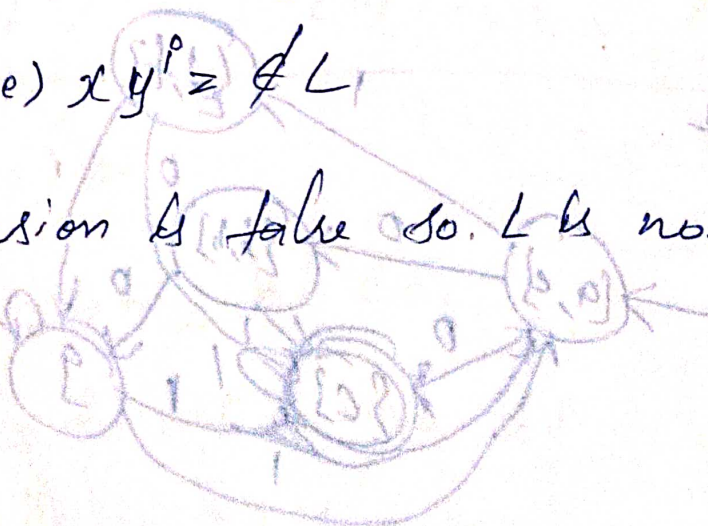
Such that $i \geq 0$.

4. Assume the string xy^iz where

$i \geq 0$ (i.e) $xy^iz \notin L$

5. Assumption is false so L is not

regular.



Problem 1

1. Prove that $L = \{0^n 1^n\}$ is not regular.

Solution:

$$L = \{0^1, 0011, 000111, \dots\}$$

1. Assume L is regular and $n = 2$

2. $w = 0011$, $|w| = 4$ $|w| \geq n$ $4 \geq 2$ — (1)

$$w = xyz \rightarrow \begin{matrix} 0 & 0 & 11 \\ x & y & z \end{matrix}$$

$$|xy| \leq n \Rightarrow 2 \leq 2$$

$$|y| > 0 = 1 > 0$$

3. $xy^i z$, $i = 2 \geq 0$

4. $0011 \Rightarrow \begin{matrix} x & y^2 & z \\ 0 & 00 & 11 \end{matrix}$
 $00011 \notin L$

5. Assumption is false, so L is not regular.

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2. $L = \{0^n, n \text{ is prime}\}$

3. $L = \{0^{i^2} ; i \geq 1\}$

4. $L = \{0^n ; n \text{ is perfect cube}\}$

5. $L = \{a^{n!} ; n \geq 0\}$

6. $L = \{a^{2^n} ; n \geq 1\}$